

Xudong Zhu

Ph.D. Student in Computer Science · The Ohio State University · Columbus, OH

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My research examines how large language models represent concepts and how these representations evolve during generation and training. I focus on the linear structure of these representations and its role in understanding and steering model behavior, as well as how different training methods, such as SFT and RL, reshape these learned representations.

EDUCATION

The Ohio State University

Ph.D. in Computer Science and Engineering, advised by Prof. Zhihui Zhu

2024 – Present

Columbus, OH

University of Electronic Science and Technology of China

B.S. in Computer Science · GPA 3.98 / 4.00

2020 – 2024

Chengdu, China

RESEARCH EXPERIENCE

- **Concept representations.** Study how large language models encode concepts, with a focus on linear structure, bidirectional features, and interpretable representations.
- **Generation dynamics.** Investigate how concept representations evolve during generation and how steering interventions propagate and modulate model behavior. Ongoing: Prefix Steering.
- **Training dynamics.** Investigate how training methods such as SFT and RL reshape concept representations and their structure. Ongoing: PostDyn.

PUBLICATIONS

AbsTopK: Rethinking Sparse Autoencoders for Bidirectional Features

Xudong Zhu, Mohammad Mahdi Khalili, Zhihui Zhu

[ICLR 2026](#) · [Paper](#) · [Code](#) · [Project Page](#)

From Emergence to Control: Probing and Modulating Self-Reflection in Language Models

Xudong Zhu, Jiachen Jiang, Mohammad Mahdi Khalili, Zhihui Zhu

[TMLR 2026](#) · [Paper](#) · [Code](#) · [Project Page](#)

Alleviating Subgraph-Induced Oversmoothing in Link Prediction via Coarse Graining

Xudong Zhu, Dong Hao, Ziqin Gao, Liming Pan

[Neurocomputing, 2025](#) · [Paper](#) · [Code](#) · [Project Page](#)

FCDS: Fusing Constituency and Dependency Syntax into Document-Level Relation Extraction

Xudong Zhu, Zhao Kang, Bei Hui

[LREC-COLING 2024](#) · [Paper](#) · [Code](#) · [Project Page](#)

TEACHING EXPERIENCE

Teaching Assistant, CSE 3521: Introduction to Artificial Intelligence

Autumn 2025

ACADEMIC SERVICE

Reviewer: NeurIPS, ICLR, ICML

2026

HONORS & RESEARCH SUPPORT

MMLS 2026 Travel Award, Midwest Machine Learning Symposium

2026

NAIRR Pilot Project NAIRR260106 (contributor), National AI Research Resource Pilot

Geometry and Training Dynamics of Representations in Large Language Models

2026

OPEN SOURCE

LLMUsage · Local-first LLM subscription usage monitor for Hyperland

TypeScript